DAY 30 | PRE-READ

Quality Checks Before Every Release

A light read before Day 30. About 15 minutes, no code required.

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Why read this?

Yesterday you measured how good an AI system is. Tomorrow you make those measurements automatic, so that every change to a prompt or model is checked before it reaches users.

1. The answer key

Teachers mark exams against an answer key that experts have checked. A **golden dataset** is the answer key for an AI system: questions with verified answers, kept safe and updated when policies change.

2. Small changes, surprising effects

Changing one word in a prompt, or switching to a cheaper model to save money, can quietly break answers that worked yesterday. **Regression testing** reruns the golden questions after every change and flags anything that got worse, just like the tests you wrote for code.

> **Key idea:** treat prompts like code. Version them, test them, and review changes.

3. Did we fetch the right pages?

Measure	Question	Everyday version
Precision	How much of what we fetched was useful?	Of 10 books brought, how many were relevant?
Recall	How much of what we needed did we fetch?	Of the 3 books needed, how many arrived?

4. Counting matching words

Some metrics, like **BLEU**, score an answer by how many word sequences it shares with a reference answer. They work well for translation, but can unfairly punish a correct answer worded differently. Choosing the right metric for the task matters.

5. A gate before release

In the CI pipelines you built in Week 4, tests block broken code. Tomorrow you add quality checks to that gate, so a change that makes answers worse cannot be released by accident.

6. Think about these before class

- Who in your organisation could verify golden answers for an HR or IT assistant?
- What change might someone make to save cost that could quietly reduce quality?
- Can you think of two correct answers to a question that share almost no words?

Kihon (基本): the foundation, the basics